PART IIA, PAPER 3, SUPERVISION 1
OLEG I. KITOV, UNIVERSITY OF CAMBRIDGE

SOLUTIONS TO PROBLEMS

S1.1 Inequality for all: knocking on the glass ceiling.
The data file `wagefull.dta` contains data on hourly wages and other variables for 3296 working individuals. The variables which you need to use for this exercise are

`wage`	the individual's wage (in \$ per hour)
`male`	a 0-1 dummy variable, 1 for males

(a) Draw a histogram of wages for the whole population using the `STATA` command `hist x, bin(25)`, where `x` is the variable name and `bin` says how many bins you wish to see in the histogram. What kind of distribution do you find?

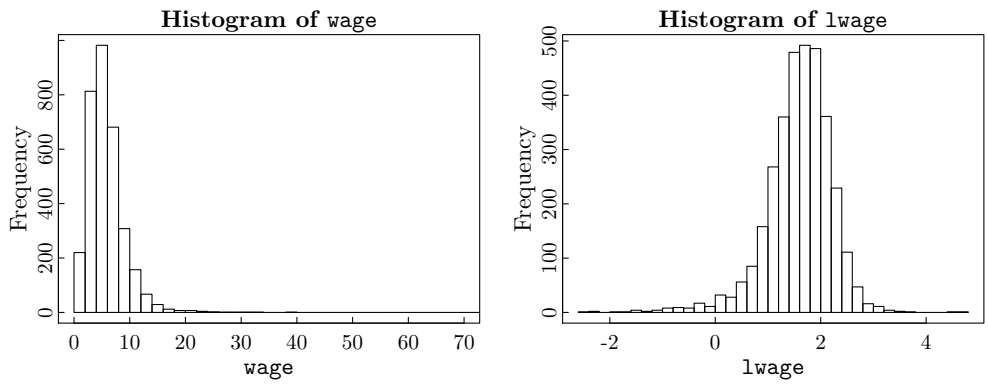

Figure 1: Histograms of $\texttt{wage}_i$ and $\texttt{lwage}_i$

(b) Create a new variable, the natural logarithm of wage, $\ln(\texttt{wage})$. Call it `lwage`. Draw a histogram of log wages. How does it compare with the histogram of wages from (a)? Why is that the case? Comment on your findings with reference to the possible assumptions about the population distribution of wages and log wages. [Hint: think about what the function $y = \ln x$ looks like.]

(c) Suggest and estimate a linear regression to compute the mean of `wage` and its standard error. Do the same for `lwage`. Explain why the log of mean wage is not equal to the mean of log wages, in other words why $\mathbb{E}[\ln(\texttt{wage})] \neq \ln(\mathbb{E}[\texttt{wage}])$.

Solution: We want to estimate

$$\texttt{wage}_i = \delta_0 + u_i, \tag{1}$$

$$\texttt{lwage}_i = \beta_0 + v_i, \tag{2}$$

where $\mathbb{E}[\texttt{wage}_i] = \delta_0$ and $\mathbb{E}[\texttt{lwage}_i] = \beta_0$, are the population means. We can use OLS estimators of δ_0 and β_0 to obtain estimates for the population log wage means:

$$\hat{\delta}_0 = \bar{\texttt{wage}} = \frac{1}{n}\sum_{i=1}^{n} \texttt{wage}_i, \qquad \hat{\beta}_0 = \bar{\texttt{lwage}} = \frac{1}{n}\sum_{i=1}^{n} \texttt{lwage}_i.$$

Now, $\mathbb{E}[\ln(\texttt{wage})] \neq \ln(\mathbb{E}[\texttt{wage}])$ because of the Jensen's inequality. Namely, for any concave function $g(\cdot)$ and a random variable X,

$$\mathbb{E}[g(X)] \leq g(\mathbb{E}[X]) \tag{3}$$

Since $\ln(\cdot)$ is a concave function and $\texttt{wage}_i$ is a random variables, we conclude that

$$\mathbb{E}[\ln(\texttt{wage})] \leq \ln(\mathbb{E}[\texttt{wage}]).$$

This is useful for later, since we can now see that testing $H_0\colon \ln(\mathbb{E}[\texttt{wage}]) = \mu_0$ is not equivalent to testing $H_0\colon \mathbb{E}[\ln(\texttt{wage})] = \mu_0$.

(d) Compute sample means and variances of `lwage` for males and females separately.

Solution: Denote $\texttt{lwage}_i$ for males by $X_i \sim \text{iid}\, N\left(\mu_X, \sigma_X^2\right)$, for $i = 1, \ldots, n_X$ with $n_X = 1727$. Denote $\texttt{lwage}_i$ for females by $Y_i \sim \text{iid}\, N\left(\mu_Y, \sigma_Y^2\right)$, for $i = 1, \ldots, n_Y$ with $n_Y = 1569$. The corresponding sample means and variance will be given by

$$\bar{X} = \frac{1}{n_X}\sum_{i=1}^{n_X} X_i, \qquad S_X^2 = \frac{1}{n_X - 1}\sum_{i=1}^{n_X}\left(X_i - \bar{X}\right)^2,$$
$$\bar{Y} = \frac{1}{n_Y}\sum_{i=1}^{n_Y} Y_i, \qquad S_Y^2 = \frac{1}{n_Y - 1}\sum_{i=1}^{n_Y}\left(Y_i - \bar{Y}\right)^2.$$

The corresponding sample estimates for men are then given by $\bar{x} = 1.693$, $s_X^2 = 0.366$, and for women by $\bar{y} = 1.475$ and $s_Y^2 = 0.398$. We can run the following code to obtain these results in `STATA`:

```
>> summ lwage if male == 1
>> summ lwage if male == 0
```

(e) State clearly the sampling distribution of the sample mean (the estimator of the population mean). Under what assumption does this sampling distribution hold? Test the hypothesis that the mean of log wages for males is equal to 1.7.

Solution: In large sample, due to the central limit theorem, the estimator of the population mean has the following (asymptotic) sampling distribution:

$$\bar{X} \overset{a}{\sim} N\left(\mu_X, \frac{\sigma_X^2}{n_X}\right) \iff \frac{\bar{X} - \mu_X}{\sqrt{\sigma_X^2/n_X}} \overset{a}{\sim} N(0,1). \tag{4}$$

We do not need any additional assumptions to derive this sampling distribution. However, in part (d) we assumed that the log wages are normally distributed for men and women, and hence, it would be correct to say that the exact distribution, where S_X^2 is the unbiased estimator of σ_X^2 is

$$\frac{\bar{X} - \mu_X}{\sqrt{S_X^2/n_X}} \sim t_{n_X - 1}.$$

For practical purposes, since $t_n \to N(0,1)$, as $n \to \infty$, we can still refer to the sampling distribution in (4). We test $H_0\colon \mu_X = \mu_X^0 = 1.7$ against $H_1\colon \mu_X \neq 1.7$.

Pick significance level $\alpha = 0.01$. Recall that under the assumption that the null is true, $\bar{X}$ has the following sampling distribution:

$$T_{\mu_X} = \frac{\bar{X} - \mu_X^0}{\sqrt{S_X^2/n_X}} \overset{a}{\sim} N(0,1). \tag{5}$$

Reject the null if the observed sample test statistic is $|t_{\mu_X}| > z_{0.995} = 2.576$, where $z_{0.995}$ is such that $\Phi(z_{0.995}) = 0.995$. Compute the test statistic,

$$t_{\mu_X} = \frac{\bar{x} - \mu_X^0}{\sqrt{s_X^2/n_X}} = \frac{1.693 - 1.7}{\sqrt{0.366/1727}} = -0.48.$$

We cannot reject the null at $\alpha = 0.01$.

(f) Explain how you would use the estimates from (d) to conduct a test of the hypothesis that the mean of log wages is the same for males and females. State the sampling distribution of the relevant statistic, and conduct such a test.

Solution: Suppose the wages for males and females are independent. We want to test for equality of the population means, $H_0\colon \mu_X = \mu_Y$ against $H_1\colon \mu_X \neq \mu_Y$. For inference, use the difference between the sample means, $\bar{D} = \bar{X} - \bar{Y}$, which are distributed as $\bar{X} \sim N\left(\mu_X, \sigma_X^2/n_X\right)$ and $\bar{Y} \sim N\left(\mu_Y, \sigma_Y^2/n_Y\right)$, respectively, according to the CLT. Clearly, $\mathbb{E}\left[\bar{X} - \bar{Y}\right] = \mu_X - \mu_Y$. The population variances are unknown and so:

$$\bar{D} = \bar{X} - \bar{Y} \overset{a}{\sim} N\left(\mu_X - \mu_Y, \frac{S_X^2}{n_X} + \frac{S_Y^2}{n_Y}\right). \tag{6}$$

Equation (6) applies when variances are unknown, and unbiased estimators are used:

$$S_X^2 = \frac{1}{n_X - 1}\sum_{i=1}^{n_X}\left(X_i - \bar{X}\right)^2, \qquad S_Y^2 = \frac{1}{n_Y - 1}\sum_{i=1}^{n_Y}\left(Y_i - \bar{Y}\right)^2. \tag{7}$$

To derive the variance of $\bar{D}$, observe that

$$\begin{aligned}
\operatorname{Var}\left(\bar{X} - \bar{Y}\right) &= \operatorname{Var}\left(\frac{1}{n_X}\sum_{i=1}^{n_X} X_i + \frac{1}{n_Y}\sum_{i=1}^{n_Y} Y_i\right) \\
[\text{independence of } X_i \text{ and } Y_i] &= \operatorname{Var}\left(\frac{1}{n_X}\sum_{i=1}^{n_X} X_i\right) + \operatorname{Var}\left(\frac{1}{n_Y}\sum_{i=1}^{n_Y} Y_i\right) \\
&= \left(\frac{1}{n_X}\right)^2 \operatorname{Var}\left(\sum_{i=1}^{n_X} X_i\right) + \left(\frac{1}{n_Y}\right)^2 \operatorname{Var}\left(\sum_{i=1}^{n_Y} Y_i\right) \\
[X_i \text{are iid and } Y_i \text{ are iid}] &= \left(\frac{1}{n_X}\right)^2 \sum_{i=1}^{n_X} \operatorname{Var}(X_i) + \left(\frac{1}{n_Y}\right)^2 \sum_{i=1}^{n_Y} \operatorname{Var}(Y_i) \\
&= \frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}.
\end{aligned}$$

The test statistic under H_0, i.e. when $\mu_X - \mu_Y = 0$, is takes one of the following two forms, depending on whether variances are known:

$$T_D = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_X^2}{n_X} + \frac{S_Y^2}{n_Y}}} \overset{a}{\sim} N(0,1). \tag{8}$$

Reject the null at significance level α if the observed sample test statistic t is such that $|t| > z_{1-\frac{\alpha}{2}}$, where $z_{1-\frac{\alpha}{2}} = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$ is the critical value from the standard normal.

$$t_D = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} = \frac{1.693 - 1.475}{\sqrt{\frac{0.366}{1726} + \frac{0.398}{1568}}} = 10.01$$

Since $t_D > z_{0.995} = 2.576$, we reject the null and conclude that log wages differ for men and women.

(g) Suggest and estimate a linear regression to compute the gender log wage gap (the difference between the mean log wage for females and the mean log wage for males). Test whether the gender log wage gap is significantly different from zero. How does this test compare with your test in (f)?

Solution: We can estimate the log wage gap using

$$\texttt{lwage}_i = \beta_0 + \beta_1 \texttt{male}_i + u_i, \quad i = 1, \ldots, n. \tag{9}$$

Note that the parameter β_1 is the wage gap, since

$$\mathbb{E}\left[\texttt{lwage}_i \,|\, \texttt{male}_i = 0\right] = \beta_0,$$
$$\mathbb{E}\left[\texttt{lwage}_i \,|\, \texttt{male}_i = 1\right] = \beta_0 + \beta_1.$$

Hence, we can run this regression to compute the estimate of β_1, use this `STATA` code

```
>> reg lwage male
```

The estimate is given by $\hat{\beta}_1 = 0.218$ with the corresponding test statistic for $H_0\colon \beta_1 = 0$ given by $t_{\beta_1} = 10.13$, suggesting that it is significant. Notice that $\hat{\beta}_1$ and t_{β_1} are identical to those we computed using the two sample method in (f). The scatter plot is given in the Figure below.

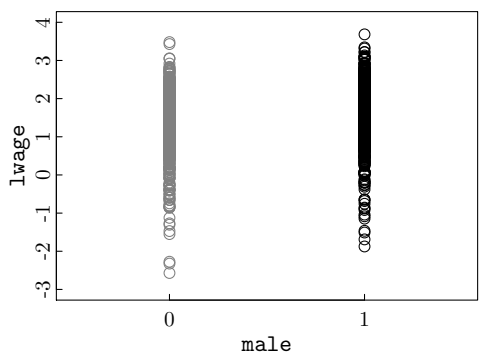

Figure 2: Scatter plot of $\texttt{male}_i$ against $\texttt{lwage}_i$

(h) State clearly the sampling distribution of the estimator of the population variance.

Solution: Recall the definition of the χ^2-distribution: given $Z_i \sim \text{iid}\, N(0,1)$, for $i = 1, \ldots, n$, the random variable U with a chi-squared distribution with n degrees of freedom is defined as

$$U = \sum_{i=1}^{n} Z_i^2 \sim \chi_n^2. \tag{10}$$

Use the definition of a χ^2-distributed random variable to figure out the sampling distribution of the variance estimators. From the single-sample hypothesis testing about the population variance, $H_0 \colon \sigma_X^2 = \sigma_0^2$, for some hypothesized value σ_0^2, we know that under the null hypothesis, the variance estimators follow a χ^2-distribution:

$$\frac{(n_X - 1)\, S_X^2}{\sigma_0^2} \sim \chi_{n_X - 1}^2, \qquad \frac{(n_Y - 1)\, S_Y^2}{\sigma_0^2} \sim \chi_{n_Y - 1}^2. \tag{11}$$

Note 1.1. Testing equality of variances using the F-test.
Consider two independent normally distributed samples (sequences) of random variables, $X_1, \ldots, X_n$, and $Y_1, \ldots, Y_m$, with unknown population means μ_x, μ_y variances σ_x^2, σ_y^2, and sample size n and m, respectively. The unbiased estimators of the population means are given by the respective sample means

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i, \qquad \bar{Y}_m = \frac{1}{m} \sum_{i=1}^{m} Y_i. \tag{12}$$

The unbiased estimators of the population variances are then given by

$$S_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}_n\right)^2, \quad S_y^2 = \frac{1}{m-1} \sum_{i=1}^{m} \left(Y_i - \bar{Y}_m\right)^2. \tag{13}$$

such that $\mathbb{E}\left[S_x^2\right] = \sigma_x^2$ and $\mathbb{E}\left[S_y^2\right] = \sigma_y^2$. Note that in (13) one degree of freedom is lost and we divide by the sample size minus one. This is because the unbiased estimator of the variance in (13) is a function of the sample mean, which in turn is itself estimated from the data, this way $\mathbb{E}\left[S_x^2\right] = \sigma_x^2$ and $\mathbb{E}\left[S_y^2\right] = \sigma_y^2$.

Definition 1.1. (χ^2-distribution). Suppose $Z_i \sim \text{iid}\, N(0,1)$, for $i = 1, \ldots, n$. The random variable U with a chi-squared distribution with n degrees of freedom is defined as

$$U = \sum_{i=1}^{n} Z_i^2 \sim \chi_n^2. \tag{14}$$

Use the definition of a χ^2-distributed random variable to figure out the sampling distribution of the variance estimators from (13). From the single-sample hypothesis testing about the population variance, $H_0 \colon \sigma_x^2 = \sigma_0^2$, for some hypothesized value σ_0^2, we know that under the null hypothesis, the variance estimators follow

a χ^2-distribution:

$$\frac{(n-1)\,S_x^2}{\sigma_0^2} \sim \chi^2_{n-1}, \qquad \frac{(m-1)\,S_y^2}{\sigma_0^2} \sim \chi^2_{m-1}. \tag{15}$$

What we are ultimately interested in, is to test a two-sample hypothesis that the population variances are the same: $H_0\colon \sigma_x^2 = \sigma_y^2$ against $H_1\colon \sigma_x^2 \neq \sigma_y^2$. Here we will show how to derive the F-test statistic for this hypothesis. First, let's define the F distribution.

Definition 1.2. (F-distribution). If two independent random variables, U_1 and U_2, each have a chi-squared distribution with d_1 and d_2 degrees of freedom, respectively, $U_1 \sim \chi^2_{d_1}$ and $U_2 \sim \chi^2_{d_2}$, then the following random variable V has an F_{d_1,d_2} distribution:

$$V = \frac{U_1/d_1}{U_2/d_2} \sim F_{d_1,d_2}. \tag{16}$$

We can say that the F_{d_1,d_2} distribution comes about as a quotient of two independent chi-squared distributions with d_1 and d_2 degrees of freedom, each corrected for their respective degrees of freedom. Figure 3 shows the pdf and the cdf of several F_{d_1,d_2} distributed random variables, namely $F_{20.5}$, $F_{50,50}$ and $F_{2,50}$. Note the unlike the normal distribution, the density function of the F distribution is asymmetric. Moreover, it is only defined for non-negative values, $x \geq 0$. This feature is important for constructing confidence intervals and conducting inference on the population variance σ^2.

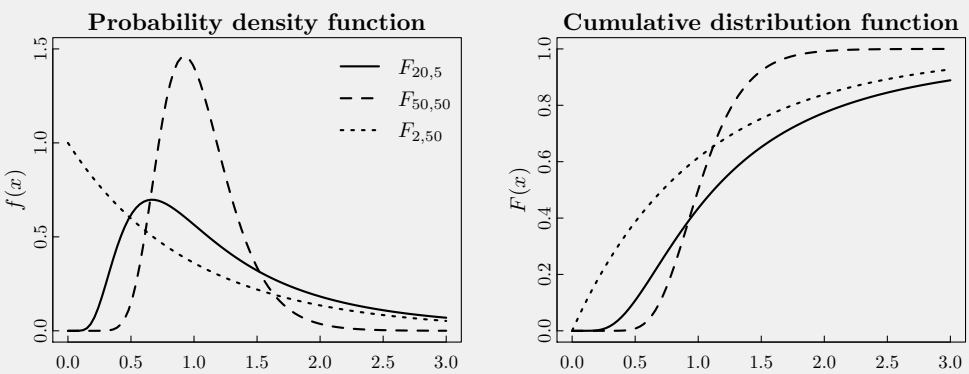

Figure 3: The pdf and the cdf of the F_{d_1,d_2}-distribution

How do we now apply this definition of the F-distributed random variable V to test $H_0\colon \sigma_x^2 = \sigma_y^2$? Observe that under H_0 the population variances are equal to the same number, $\sigma_x^2 = \sigma_y^2 = \sigma^2$. Now, let's denote our random variables U_1 and U_2 in the definition of the F-distribution (16) to be

$$U_1 = \frac{(n-1)\,S_x^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad U_2 = \frac{(m-1)\,S_y^2}{\sigma^2} \sim \chi^2_{m-1}. \tag{17}$$

Observe that the chi-squared distributed random variable U_1 has $d_1 = n - 1$ degrees of freedom, while U_2 has $d_2 = m - 1$ degrees of freedom. We need to account for that when using definition (16) to construct the F-test statistic. Furthermore, note that in the above, we used the fact that under the null, the population variances are the same and equal to σ^2. The result for the F-test statistic follows by applying the definition of the F-distributed random variable from (16) to U_1 and U_2 in (17):

$$V = \frac{U_1/(n-1)}{U_2/(m-1)} = \left[\frac{(n-1)\,S_x^2}{\sigma^2\,(n-1)}\right] / \left[\frac{(m-1)\,S_y^2}{\sigma^2\,(m-1)}\right] = \frac{S_x^2}{S_y^2} \sim F_{n-1,m-1}. \tag{18}$$

What we then need to do is to compute the sample F-test statistic as the quotient of the variance estimators, S_x^2/S_y^2, and compare it to the critical values of the $F_{n-1,m-1}$ distribution. For the two-sided alternative, $H_1 : \sigma_x^2 \neq \sigma_y^2$, we reject H_0 at the significance level α if the sample test statistic F is such that $F < F_{\frac{\alpha}{2},n-1,m-1}$ or $F > F_{1-\frac{\alpha}{2},n-1,m-1}$. We distributed the overall rejection probability α equally in both tails of the distribution. Clearly, for the two-sided alternative, we need to look up two critical values from the statistical tables for the $F_{n-1,m-1}$ distribution, namely $F_{\frac{\alpha}{2},n-1,m-1}$ and $F_{1-\frac{\alpha}{2},n-1,m-1}$. This is because the density of F is non-symmetrical and is only defined for non-negative values, $x \geq 0$ (recall Figure 3).

We can simplify the testing procedure somewhat. Since variances are strictly positive, one of the variance estimators will definitely be larger than the other. Consequently, it is much more common to test H_0 against a one-sided alternative. The rule of thumb is to look at the sample variance that is the largest of the two. Suppose, for instance, that $S_x^2 > S_y^2$, and formulate the alternative as the corresponding population variance also being larger, $H_1 \colon \sigma_x^2 > \sigma_y^2$. In this case, we only need to look at the right tail of the $F_{n-1,m-1}$ distribution and compare our F-test statistic S_x^2/S_y^2 to a single critical value $F_{1-\alpha,n-1,m-1}$, rejecting H_0 if it is exceeded.

(i) Explain how you would test the hypothesis that the variance of log wages is the same for males and females, stating the sampling distribution of the relevant statistic. Conduct such a test using the estimates of variances you computed in (d). Does the test confirm your intuition about which variance should be larger?

Solution: We test $H_0 \colon \sigma_X^2 = \sigma_Y^2$ against $H_1 \colon \sigma_X^2 > \sigma_Y^2$, using the F-statistic

$$V = \frac{S_X^2}{S_Y^2} \sim F_{n_X-1,n_Y-1}.$$

In this case, we only need to look at the right tail of the F_{n_X-1,n_Y-1} distribution and compare our F-test statistic S_X^2/S_Y^2 to a single critical value $F_{1-\alpha,n_X-1,n_Y-1}$, rejecting H_0 if it is exceeded. For out data, the observed sample test statistic is

$$v = \frac{s_Y^2}{s_X^2} = \frac{0.398}{0.366} = 1.087.$$

The critical value for $\alpha = 0.01$ is given by $F_{0.99,1568,1726} = 1.1215$ and we cannot

reject the null at this significance level. However, since $F_{0.95,1568,1726} = 1.084$, we can (marginally) reject the null at $\alpha = 0.05$.

S1.2 The OLS estimators are not just better, they are the best!
A set of observations y_i is generated by the following model

$$y_i = \beta x_i + \varepsilon_i, \quad i = 1, \ldots, n \tag{19}$$

with x_i a sequence of fixed numbers, and ε_i a set of iid random variables with $\mathbb{E}[\varepsilon_i] = 0$ and $\mathbb{E}\left[\varepsilon_i^2\right] = \sigma^2$, and β is an unknown parameter. Using the observations $\{y_i, x_i\}$, $i = 1, \ldots, n$, three estimators for β are considered:

$$\hat{\beta}^{(1)} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}, \qquad \hat{\beta}^{(2)} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}, \qquad \hat{\beta}^{(3)} = \sum_{i=1}^n \left(\frac{y_i}{x_i}\right). \tag{20}$$

Express the three estimators in terms of x_i, ε_i, $i = 1, \ldots, n$, and β. Hence, find the expectations and variances of the estimators. Based on these calculations, which estimator is most preferred? Explain why.

Solution: Equation (19) is a linear regression without an intercept. First, observe that since $\mathbb{E}[\varepsilon_i] = 0$, we can express the variance of the error term as

$$\mathrm{Var}(\varepsilon_i) = \mathbb{E}\left[\varepsilon_i^2\right] - \left(\mathbb{E}[\varepsilon_i]\right)^2 = \mathbb{E}\left[\varepsilon_i^2\right] = \sigma^2. \tag{21}$$

The distribution of the error terms is then given by $\varepsilon_i \sim \text{iid}\left(0, \sigma^2\right)$, $i = 1, \ldots, n$. This question emphasized the fact that the estimators are random variables and so have probability distributions. Recall that an estimator is a rule for computing the *best guess* for the value of the population parameter, given any data sample. For some observed (realized) data sample, the estimator takes a particular value, which is called the estimate. You can then think of estimates as draws (realizations) from the distribution of the estimator: one draw corresponds to one observed sample, which correspond to one estimate.

Unbiased estimators are such whose distributions are centered around the true parameter, in our case β. Efficient estimators are such whose distributions are more densely concentrated, that is the variance is low and the distribution is narrower (but does not have to be centered around the true parameter). If the estimator is both unbiased and efficient, then its distribution is narrow and centered around the true parameter. In this case, it is very likely that for any possible observed random sample, the value of the estimate from that sample will be very close to the population parameter that it estimates. When working with estimators and their properties, it is always a good idea to express the estimators in terms of the true parameter and the error termsε_i. For this, substitute for y_i from (19) into the estimators:

$$\hat{\beta}^{(1)} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \frac{\sum_{i=1}^n x_i\left(x_i\beta + \varepsilon_i\right)}{\sum_{i=1}^n x_i^2} = \beta\left(\frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2}\right) + \frac{\sum_{i=1}^n x_i\varepsilon_i}{\sum_{i=1}^n x_i^2} = \beta + \frac{\sum_{i=1}^n x_i\varepsilon_i}{\sum_{i=1}^n x_i^2}.$$

$$\hat{\beta}^{(2)} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i} = \frac{\sum_{i=1}^n \left(x_i\beta + \varepsilon_i\right)}{\sum_{i=1}^n x_i} = \beta\left(\frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n x_i}\right) + \frac{\sum_{i=1}^n \varepsilon_i}{\sum_{i=1}^n x_i} = \beta + \frac{\sum_{i=1}^n \varepsilon_i}{\sum_{i=1}^n x_i}.$$

$$\hat{\beta}^{(3)} = \sum_{i=1}^n \left(\frac{y_i}{x_i}\right) = \sum_{i=1}^n \left(\frac{x_i\beta + \varepsilon_i}{x_i}\right) = \sum_{i=1}^n \beta\left(\frac{x_i}{x_i}\right) + \sum_{i=1}^n \left(\frac{\varepsilon_i}{x_i}\right) = n\beta + \sum_{i=1}^n \left(\frac{\varepsilon_i}{x_i}\right).$$

Computing expectations of the estimators is now very easy if we note that x_i are fixed numbers that can be taken outside the expectation operator as constants, and that the error terms have zero mean, $\mathbb{E}[\varepsilon_i] = 0$, for all i.

$$\mathbb{E}\left[\hat{\beta}^{(1)}\right] = \mathbb{E}\left[\beta + \frac{\sum_{i=1}^n x_i\varepsilon_i}{\sum_{i=1}^n x_i^2}\right] = \beta + \frac{\sum_{i=1}^n x_i\mathbb{E}[\varepsilon_i]}{\sum_{i=1}^n x_i^2} = \beta. \tag{22}$$

$$\mathbb{E}\left[\hat{\beta}^{(2)}\right] = \mathbb{E}\left[\beta + \frac{\sum_{i=1}^n \varepsilon_i}{\sum_{i=1}^n x_i}\right] = \beta + \frac{\sum_{i=1}^n \mathbb{E}[\varepsilon_i]}{\sum_{i=1}^n x_i} = \beta. \tag{23}$$

$$\mathbb{E}\left[\hat{\beta}^{(3)}\right] = \mathbb{E}\left[n\beta + \sum_{i=1}^n \left(\frac{\varepsilon_i}{x_i}\right)\right] = n\beta + \sum_{i=1}^n \left(\frac{1}{x_i}\right)\mathbb{E}[\varepsilon_i] = n\beta. \tag{24}$$

Clearly, $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$ are unbiased, while $\hat{\beta}^{(3)}$ has a bias of $(n-1)\beta$. Now, let's compute the variances of the estimators. Keep in mind that $\mathrm{Var}(\beta) = 0$, since β is a constant. Also, recall $\mathrm{Var}(\varepsilon_i) = \sigma^2$ and $\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$, since ε_i are iid.

$$\mathrm{Var}\left(\hat{\beta}^{(1)}\right) = \mathrm{Var}\left(\beta + \frac{\sum_{i=1}^n x_i\varepsilon_i}{\sum_{i=1}^n x_i^2}\right) = \frac{\sum_{i=1}^n x_i^2}{\left(\sum_{i=1}^n x_i^2\right)^2}\mathrm{Var}(\varepsilon_i) = \left(\frac{1}{\sum_{i=1}^n x_i^2}\right)\sigma^2. \tag{25}$$

$$\mathrm{Var}\left(\hat{\beta}^{(2)}\right) = \mathrm{Var}\left(\beta + \frac{\sum_{i=1}^n \varepsilon_i}{\sum_{i=1}^n x_i}\right) = \frac{\sum_{i=1}^n \mathrm{Var}(\varepsilon_i)}{\left(\sum_{i=1}^n x_i\right)^2} = \left(\frac{n}{\left(\sum_{i=1}^n x_i\right)^2}\right)\sigma^2. \tag{26}$$

$$\mathrm{Var}\left(\hat{\beta}^{(3)}\right) = \mathrm{Var}\left(n\beta + \sum_{i=1}^n \left(\frac{\varepsilon_i}{x_i}\right)\right) = \sum_{i=1}^n \left(\frac{1}{x_i^2}\right)\mathrm{Var}(\varepsilon_i) = \left(\sum_{i=1}^n \left(\frac{1}{x_i^2}\right)\right)\sigma^2. \tag{27}$$

A detailed derivation of the variance of $\hat{\beta}^{(1)}$ is provided below:

$$\begin{aligned}
\mathrm{Var}\left(\hat{\beta}^{(1)}\right) &= \mathrm{Var}\left(\beta + \frac{\sum_{i=1}^n x_i\varepsilon_i}{\sum_{i=1}^n x_i^2}\right). \\
[\![\beta \text{ is a constant}]\!] &= \mathrm{Var}\left(\frac{\sum_{i=1}^n x_i\varepsilon_i}{\sum_{i=1}^n x_i^2}\right) \\
[\![x_i \text{ are constant}]\!] &= \left[\frac{1}{\left(\sum_{i=1}^n x_i^2\right)^2}\right]\mathrm{Var}\left(\sum_{i=1}^n x_i\varepsilon_i\right) \\
&= \left[\frac{1}{\left(\sum_{i=1}^n x_i^2\right)^2}\right]\mathrm{Var}\left(x_1\varepsilon_1 + x_2\varepsilon_2 + \cdots + x_n\varepsilon_n\right) \\
[\![\text{variance of a sum}]\!] &= \left[\frac{1}{\left(\sum_{i=1}^n x_i^2\right)^2}\right]\left[\sum_{i=1}^n \mathrm{Var}(x_i\varepsilon_i) + 2\sum_{i\neq j}\mathrm{Cov}(x_i\varepsilon_i, x_j\varepsilon_j)\right] \\
[\![\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0, i \neq j]\!] &= \left[\frac{1}{\left(\sum_{i=1}^n x_i^2\right)^2}\right]\sum_{i=1}^n \mathrm{Var}(x_i\varepsilon_i) \\
[\![x_i \text{ are constant}]\!] &= \left[\frac{1}{\left(\sum_{i=1}^n x_i^2\right)^2}\right]\sum_{i=1}^n x_i^2\mathrm{Var}(\varepsilon_i).
\end{aligned}$$

Finally, since $\operatorname{Var}(\varepsilon_i) = \sigma^2$, this can be simplified as:

$$\operatorname{Var}\left(\hat{\beta}^{(1)}\right) = \left[\frac{\sum_{i=1}^n x_i^2}{\left(\sum_{i=1}^n x_i^2\right)^2}\right] \operatorname{Var}(\varepsilon_i) = \left(\frac{1}{\sum_{i=1}^n x_i^2}\right)\sigma^2.$$

Now, since $\hat{\beta}^{(3)}$ is biased, we only need to compare variances of $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$, as they are preferred to $\hat{\beta}^{(3)}$. Intuitively, $\hat{\beta}^{(1)}$ should be preferred, since it the OLS estimator in a regression without an intercept and so according to the Gauss-Markov theorem has the lowest variance among all unbiased estimators of β. We can show that $\hat{\beta}^{(1)}$is the most efficient estimator out of the two unbiased estimators directly comparing the variances:

$$\operatorname{Var}\left(\hat{\beta}^{(1)}\right) = \frac{\sigma^2}{\sum_{i=1}^n x_i^2} \leq \frac{n\sigma^2}{\left(\sum_{i=1}^n x_i\right)^2} = \operatorname{Var}\left(\hat{\beta}^{(2)}\right).$$

In order to show the inequality in the middle, we need to prove the following statement: $\left(\sum_{i=1}^n x_i\right)^2 \leq n\sum_{i=1}^n x_i^2$. For that we will use the Cauchy-Schwartz inequality. First, consider the following inequality for two sequences of real numbers, x_i and y_i, for $i = 1, \ldots, n$:

$$\sum_{i=1}^n (tx_i + y_i)(tx_i + y_i) \geq 0,$$

where t could be any real value. This summation is clearly greater than zero, since for all i, $(tx_i + y_i)(tx_i + y_i) = (tx_i + y_i)^2 \geq 0$. By expanding the brackets we have

$$\left(\sum_{i=1}^n x_i^2\right) t^2 + \left(2\sum_{i=1}^n x_i y_i\right) t + \sum_{i=1}^n y_i^2 \geq 0.$$

This equation is a quadratic equation in t, such that $at^2 + bt + c \geq 0$, where the coefficients are given by

$$a = \sum_{i=1}^n x_i^2, \qquad b = 2\sum_{i=1}^n x_i y_i, \qquad c = \sum_{i=1}^n y_i^2.$$

Since the inequality is non-negative, we must have $b^2 \leq 4ac$. Substituting the values of a, b, c, then we have

$$\left(\sum_{i=1}^n 2x_i y_i\right)^2 \leq 4\sum_{i=1}^n y_i^2 \sum_{i=1}^n x_i^2 \implies \left(\sum_{i=1}^n x_i y_i\right)^2 \leq \sum_{i=1}^n y_i^2 \sum_{i=1}^n x_i^2. \tag{28}$$

This is a Cauchy-Schwartz inequality. Finally, if we assume y_i to take the value of 1 for all $i = 1, \ldots, n$, then the Cauchy-Schwartz inequality implies that

$$\left(\sum_{i=1}^n x_i\right)^2 \leq n\sum_{i=1}^n x_i^2.$$

Simulation 1.1. Comparing distributions of unbiased estimators.

Let's illustrate the difference between the two unbiased estimators from this question graphically by plotting the distributions of $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$ for a hypothetical dataset. Recall that

$$\hat{\beta}^{(1)} = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}, \qquad \hat{\beta}^{(2)} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i}.$$

We will use the Monte Carlo simulation method. You should make sure you understand what is happening here very carefully, as we will be using this technique to study hypothetical examples practically every week. Consider the following data generating process for y_i, $i = 1, \ldots, n$:

$$y_i = 2x_i + \varepsilon_i, \quad x_i \sim U[0, 10], \quad \varepsilon_i \sim N(0, 1). \tag{29}$$

The population parameter $\beta = 2$. The explanatory variables x_i are drawn from a continuous uniform distribution on an interval $[0, 10]$. The error terms ε_i , independent of the explanatory variables x_i for all i, are standard normal random variables. For simplicity, fix the sample size at $n = 50$. Use the random number generator to generate the explanatory variables only once and keep them fixed at $(x_1, \ldots, x_{50})$. Then generate $m = 1000$ samples of the error terms, each sample of size $n = 50$. For example, for some sample $j = 1, \ldots, 1000$, the error term for observations $i = 1, \ldots, 50$ is drawn from the standard normal distribution $\varepsilon_{j,i} \sim N(0, 1)$ to form the j'th sample of the error terms, $(\varepsilon_{j,1}, \ldots, \varepsilon_{j,n})$. Finally, use the fixed x_i and the samples of the error terms to generate $m = 1000$ samples of the dependent variables y_i using (29). For instance, for sample $j = 1, \ldots, m$, the dependent variable is given by $y_{j,i} = 2x_i + \varepsilon_{j,i}$, for all $i = 1, \ldots, 50$.

Essentially, we have 1000 data samples that are generated using the population linear model (29). The difference in the values of y_i between samples is purely driven by varying error terms between the samples. This is because we allowed the explanatory variables $(x_1, \ldots, x_{50})$ to have the same values (be fixed) for each sample. Now, let's pretend that we don't know what the β parameter is and instead estimate it from the sample data. For each sample $j = 1, \ldots, 1000$, compute the estimates of β using estimators $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$ from expressions (20) and the sample data $\{(y_{j,1}, x_1), \ldots, (y_{j,n}, x_n)\}$. Denote the estimates of β corresponding to sample j by $\hat{\beta}_j^{(1)}$ and $\hat{\beta}_j^{(2)}$, respectively. This process can be schematically illustrated in a table as follows:

Sample j	**Sample data**		**Estimators**
1	$\{(y_{1,1}, x_1), (y_{1,2}, x_2), \ldots, (y_{1,50}, x_{50})\}$	$\Longrightarrow$	$\hat{\beta}_1^{(1)}, \hat{\beta}_1^{(2)}$
2	$\{(y_{2,1}, x_1), (y_{2,2}, x_2), \ldots, (y_{2.50}, x_{50})\}$	$\Longrightarrow$	$\hat{\beta}_2^{(1)}, \hat{\beta}_2^{(2)}$
$\vdots$	$\vdots$		$\vdots$
m	$\{(y_{m,1}, x_1), (y_{m,2}, x_2), \ldots, (y_{m,50}, x_{50})\}$	$\Longrightarrow$	$\hat{\beta}_m^{(1)}, \hat{\beta}_m^{(2)}$

This means that we have $m = 1000$ estimates of β computed using $\hat{\beta}^{(1)}$ and $m = 1000$ estimates of β computed using $\hat{\beta}^{(2)}$, each corresponding to $m = 1000$ random data samples. We can now plot the sampling distributions of the estimators $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$ (think of these as smooth histograms). These are presented in Figure 4 together with a sample scatter plot for one arbitrary data sample.

Notice that the predicted (fit) lines on the left panel of the Figure generated from the two estimators, $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$, are almost indistinguishable. Both distributions (right panel) are centered around the population parameter $\beta = 2$, which is unsurprising as both estimators are unbiased. Moreover, we can see that the distribution of $\hat{\beta}^{(2)}$ is slightly wider, which is precisely the illustration of $\text{Var}(\hat{\beta}^{(2)}) \geq \text{Var}(\hat{\beta}^{(1)})$. This means that for a randomly drawn data sample, we would expect $\hat{\beta}^{(1)}$ to yield an estimate close to β more often (with higher probability) than $\hat{\beta}^{(2)}$. These are exactly the tow properties that we would like our estimators to have. This is why econometricians are obsessed with unbiased and efficient estimators - they yield estimates sufficiently close to the truth more often than estimators that do not possess these properties.

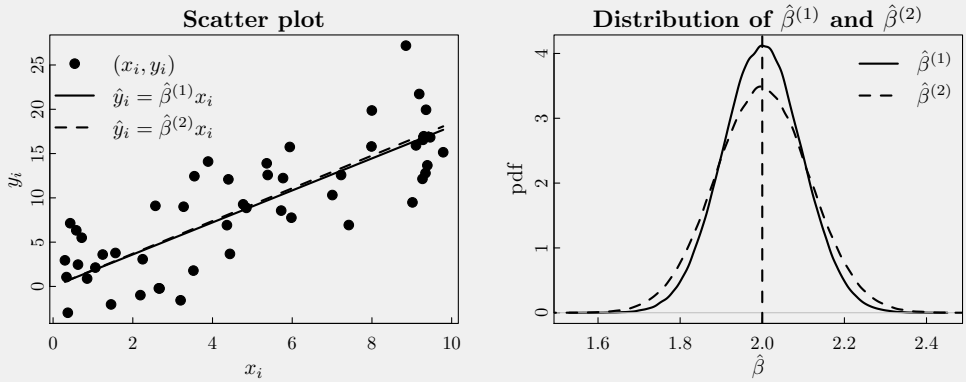

Figure 4: Scatter plot of data from (29) and the distribution of $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$

S1.3 Where regressions were born: the original regression to the mean.
Sir Francis Galton, the forefather of the linear regression method in social sciences, studied the relationship between the height of fathers (X_i) and their sons (Y_i) using the following population regression

$$Y_i = \alpha + \beta X_i + u_i, \quad i = 1, \ldots, n \tag{30}$$

where the residual u_i is an unobserved random variable with zero conditional mean, $\mathbb{E}[u_i|X_i] = 0$, and constant variance. Note that X_i and Y_i are random variables, not fixed numbers, and hence have probability distributions. A further crucial assumption is made that the distribution of height in the population is approximately the same in each generation with mean μ and variance σ^2. Consider the population regression, that is no estimation is taken place. We are interested in interpreting the population parameters in (30) and try to understand the intergenerational dynamics of height.

(a) Derive the expressions for and interpret the population parameters α and β. Compare

these with the expressions for the OLS estimators $\hat{\alpha}$ and $\hat{\beta}$. [Hint: compute $\mathbb{E}[Y_i]$ and $\text{Cov}(Y_i, X_i)$. You may use $\mathbb{E}[u_i] = 0$ and $\text{Cov}(X_i, u_i) = 0$.]

Solution: For α, take the unconditional expectation on both sides of (30):

$$\mathbb{E}[Y_i] = \mathbb{E}[\alpha + \beta X_i + u_i] = \alpha + \beta \mathrm{E}[X_i] + \mathbb{E}[u_i],$$

and use $\mathbb{E}[u_i] = 0$ to conclude that

$$\alpha = \mathbb{E}[Y_i] - \beta \mathbb{E}[X_i]. \tag{31}$$

Typically, we would interpret α as the expected height of a son whose father has height $X_i = 0$, that is $\alpha = \mathbb{E}[Y_i | X_i = 0]$. However, this clearly does not have any physical meaning and we simply say that α is the population intercept parameter. For β, take covariance with X_i on both sides of (30):

$$\text{Cov}(Y_i, X_i) = \text{Cov}(\alpha + \beta X_i + u_i, X_i) = \beta \text{Cov}(X_i, X_i) + \text{Cov}(u_i, X_i).$$

Use the fact that $\text{Cov}(X_i, X_i) = \text{Var}(X_i)$ and that $\text{Cov}(u_i, X_i) = 0$ to derive

$$\beta = \frac{\text{Cov}(Y_i, X_i)}{\text{Var}(X_i)}. \tag{32}$$

The population slope parameter β is the additional effect that a small increase in the height of the father has on the height of his son, formally $\beta = \frac{\partial Y_i}{\partial X_i}$. Note that the population parameters α and β have the same expressions as the OLS estimators $\hat{\alpha}$ and $\hat{\beta}$, where the sample moments $\bar{X}$, $\bar{Y}$, S_{xy} and S_{xx} are replaced with their population counterparts, $\mathbb{E}[X_i]$, $\mathbb{E}[Y_i]$, $\text{Cov}(X_i, Y_i)$ and $\text{Var}(X_i)$, respectively. To see this explicitly, compare (31) and (32) with the OLS estimators

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}, \qquad \hat{\beta} = \frac{\sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (X_i - \bar{X})^2} = \frac{S_{xy}}{S_{xx}}.$$

(b) What does the population regression (30) say about the factors determining the son's height? Provide intuition for the meaning of the error term u_i and the assumption of zero condition mean $\mathbb{E}[u_i | X_i] = 0$.

Solution: The population regression (30) tells us that the height of the son is a linear function of the height of his dad, plus some error term u_i, which is an unobserved random variable. The term u_i encapsulates the overall effect that all other variables, apart from father's height, have on the height of the son. As far as we are concerned, none of these other variables are observed. Then the assumption $\mathbb{E}[u_i | X_i] = 0$ means that the expected influence of these unobserved variables does not depend on the height of the father. That is, we expect that a change in the value of X_i will produce no effect on the value of u_i for all individuals, and will only affect Y_i directly.

(c) Show that β is equal to the intergenerational correlation coefficient ρ_{xy}, which is the correlation between the height of fathers and their sons, $\text{Corr}(X_i, Y_i)$.

Solution: The assumption that the distribution of heights in the population is the same for two successive generations means that $\mathbb{E}[X_i] = \mathbb{E}[Y_i] = \mu$ and $\text{Var}(X_i) = \text{Var}(Y_i) = \sigma^2$. From expression (32) for β it follows that

$$\beta = \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)} = \frac{\text{Cov}(X_i, Y_i)}{\sqrt{\text{Var}(X_i)}\sqrt{\text{Var}(X_i)}} = \frac{\text{Cov}(X_i, Y_i)}{\sqrt{\text{Var}(X_i)}\sqrt{\text{Var}(Y_i)}} = \rho_{xy}. \tag{33}$$

(d) Show that the coefficient α is equal to $(1-\rho_{xy})\,\mu$.

Solution: From the definition of the intercept coefficient α, it follows that

$$\alpha = \mathbb{E}\left[Y_i\right] - \beta \mathrm{E}\left[X_i\right] = \mu - \beta\mu = (1-\rho_{xy})\,\mu. \tag{34}$$

(e) Express the conditional expectation function, $\mathbb{E}\left[Y_i|X_i\right]$, in terms of μ and ρ_{xy}.

Solution: Substitute for Y_i from (30) into the conditional expectation function and use the the assumption of zero conditional mean $\mathbb{E}\left[u_i|X_i\right] = 0$, to show that

$$\mathbb{E}\left[Y_i|X_i\right] = \mathbb{E}\left[\alpha + \beta X_i + u_i|X_i\right] = \mathbb{E}\left[\alpha|X_i\right] + \mathbb{E}\left[\beta X_i|X_i\right] + \mathbb{E}\left[u_i|X_i\right] = \alpha + \beta X_i,$$

where the second term is $\mathbb{E}\left[\beta X_i|X_i\right] = \beta \mathrm{E}\left[X_i|X_i\right] = \beta X_i$, as quite trivially, the conditional expectation of random variable conditional on itself is simply itself, and the third term is $\mathbb{E}\left[u_i|X_i\right] = 0$ by assumption. Substitute the values of α and β we found in (b) and (c) to conclude that

$$\mathbb{E}\left[Y_i|X_i\right] = (1-\rho_{xy})\,\mu + \rho_{xy}X_i. \tag{35}$$

The conditional expectation function predicts the expected value of Y_i for any given value of X_i. For any realization x (observation in a given sample) of X_i, we can then define the function $r\left(x\right)$, which predicts the sample observations of Y_i for any x:

$$r\left(x\right) = \mathbb{E}\left[Y_i|X_i = x\right] = (1-\rho_{xy})\,\mu + \rho_{xy}x. \tag{36}$$

(f) Interpret the conditional expectation function. What does it imply about the intergenerational height dynamics? If the same methods were used to study the intergenerational dynamics of income, or educational attainment or accident rates after the introduction of speed cameras what would the implications be?

Solution: The conditional expectation function, $\mathbb{E}\left[Y_i|X_i\right]$, shows the expected height of the son conditional on the height of his father. Assuming that the correlation coefficient is positive $0 < \rho_{xy} < 1$, the expected height of a son (any hypothetical son) in equation (35) is a weighted average of the height of his father and the population mean height. The weight of the height of the father is the population correlation coefficient ρ_{xy}. The weight of the population mean is one minus the population correlation coefficient, $(1-\rho_{xy})$. As expected, the weights add up to one.

Observe that for $0 \leq \rho_{xy} \leq 1$, the son's expected height will lie between the value of his father's height and the population mean. If dad was taller than the population mean, the son can expect to be shorter than his dad, and vice versa. The son will be expected to *regress to the mean*. This is illustrated in the Figure 5 below. The blue dots are the observations (Y_i, X_i), and the upward sloping line is the conditional expectation function $r\left(x\right) = \mathbb{E}\left[Y_i|X_i = x\right]$ from (35). From the line, observe that $\mathbb{E}\left[Y_i|X_i = 60\right] \approx 65$, while $\mathbb{E}\left[Y_i|X_i = 75\right] \approx 70$. This is a numerical illustration of regression to the mean. Note that the slope of the regression line is equal to the intergenerational correlation coefficient ρ_{xy}.

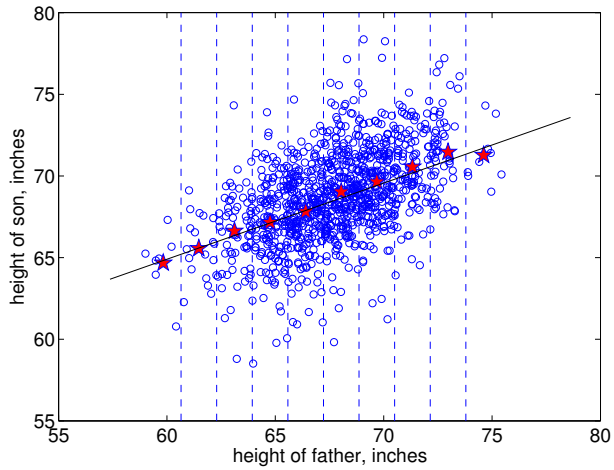

Figure 5: Regression of son's height on father's height and the CEF, $\mathbb{E}[Y_i|X_i]$

Expression in (35) comes (mechanically) from having two *stationary* distributions - distributions that do not change in time. The overall distribution of heights is assumed to be the same for all generations, which implies that the mean and the variance are time-invariant. The mean-reversion principle (regression to the mean) applies to any process with stationary distribution, e.g. income: if parents were rich, the child's income would be expected to be closer to the mean. In reality, this is of course not the case, since the average population height (as well as income and educational attainment) tends to grow with time.

(g) Consider two hypothetical cases: $\rho_{xy} = 0$ or $\rho_{xy} = 1$. Interpret the conditional expectation function and the intergenerational height dynamics for each case.

Solution: In a (totally unrealistic) hypothetical scenario when the heights of consecutive generations are completely unrelated $(\rho_{xy} = 0)$, the height of all sons would be completely unrelated from the height of their fathers. We would observe no height heritability. In fact, all sons would have the expected height exactly equal to to the population mean $\mathbb{E}[Y_i|X_i = x] = \mu$. The actual heights would of course differ, with the difference purely driven by random factors u_i, since $Y_i = \mu + u_i$. This situation is what Galton would probably call complete dominance of nurture over nature. Presumably, in this case, we would observe a lot less height variability across individuals, as everyone would be approximately the same height, μ.

At the opposite extreme $(\rho_{xy} = 1)$ the expected height of the sons would be completely determined by the height of their dads, $\mathbb{E}[Y_i|X_i = x] = x$. The actual observed height would be $Y_i = X_i$, and there is no error because correlation of 1 implies Y_i and X_i have a linear deterministic relationship. The population mean would be irrelevant for determining Y_i. In this case, we get full heritability and dominance of nature over nurture. We would observe constant historical heights within each family. A very strange situation indeed, if you think about how the first generation of humans got their heights to begin with.

S1.4 Don't let the economists drown - use the normal distribution!
Fifteen male economists are in a life raft with a maximum carrying capacity of 2850 pounds. The distribution of weights of male economists is normal with a mean of 178 pounds and with a standard deviation of 17 pounds. [Hint: define the weight of one economist be $X_i \sim N\left(178, 17^2\right)$ and work with a random variable that describes the sum of the weights, $S_{15} = \sum_{i=1}^{15} X_i$.]

(a) Find the probability that all the economists weigh less than 189 pounds.

Solution: The probability that a random economist weighs less than 189 pounds can be found using the standardization transformation of the random variable X_i into the standard normal random variable $Z_i \sim N(0,1)$ with cdf $\Phi(\cdot)$

$$\mathbb{P}(X_i \leq 189) = \mathbb{P}\left(\frac{X_i - 178}{17} \leq \frac{189 - 178}{17}\right) = \mathbb{P}\left(Z_i \leq \frac{11}{17}\right) = \Phi\left(\frac{11}{17}\right) = 0.741.$$

Since the weights of the economists are independent, we can write

$$\bigcap_{i=1}^{15} \mathbb{P}(X_i \leq 189) = \prod_{i=1}^{15} \mathbb{P}(X_i \leq 189) = [\mathbb{P}(X_i \leq 189)]^{15} = 0.741^{15} = 0.011.$$

(b) Find the probability that the raft is overloaded and will sink.

Solution: Define the sum of the weights $S_{15} = \sum_{i=1}^{15} X_i$. Then, since X_i are iid normal with mean $\mu = 178$ and variance $\sigma^2 = 17^2$, we can find

$$\mathbb{E}[S_{15}] = \mathbb{E}\left[\sum_{i=1}^{15} X_i\right] = \sum_{i=1}^{15} \mathbb{E}[X_i] = 15 \times 178 = 2670$$

$$\mathrm{Var}(S_{15}) = \mathrm{Var}\left(\sum_{i=1}^{15} X_i\right) = \sum_{i=1}^{15} \mathrm{Var}(X_i) = \sum_{i=1}^{15} \mathrm{Var}(X_i) = 15 \times 17^2 = 4335$$

where in the computation of the variance of S_{15} we use the independence of X_i and X_j for all $i \neq j$, which implies $\mathrm{Cov}(X_i, X_j) = 0$, and so the term $2\sum_{i \neq k} \mathrm{Cov}(X_i, X_j)$ is simplified to zero. We can conclude that $S_{15} \sim N(2670, 4335)$. We are looking for $\mathbb{P}(S_{15} \leq 2850)$ and can use the standardized version of S_{15}, namely

$$\mathbb{P}(S_{15} > 2850) = \mathbb{P}\left(\frac{S_{15} - 2670}{\sqrt{4335}} > \frac{2850 - 2670}{\sqrt{4335}}\right) = \mathbb{P}(Z > 2.734) = 1 - \Phi(2.734).$$

Looking for $\Phi(2.734)$ in the statistical table for the standard normal distribution, conclude that the probability of an overload is 0.003.

(c) Find the maximum number of economists that should enter the raft if the probability of overloading is not to exceed 0.0001.

Solution: We want to express the sum of weights of n economists as a function of n, and similarly to S_{15} define the general total weight:

$$S_n = \sum_{i=1}^{n} X_i \implies S_n \sim N\left(n\mu, n\sigma^2\right)$$

where $\mu = 178$ and $\sigma^2 = 17^2$, so that $S_n \sim N\left(178n, 17^2 n\right)$. We want to find n such that at the threshold of probability:

$$\mathbb{P}\left(S_n > 2850\right) = 0.0001 \iff \mathbb{P}\left(S_n \leq 2850\right) = 1 - 0.0001 = 0.9999.$$

The probability level $\alpha = 0.9999$ corresponds to $z = 3.719$, in other words $\Phi(z) = \alpha$ and $\Phi(3.719) = 0.9999$. To find $\mathbb{P}\left(S_n \leq 2850\right)$, standardize S_n:

$$\mathbb{P}\left(S_n \leq 2850\right) = \mathbb{P}\left(\frac{S_n - 178n}{17\sqrt{n}} \leq \frac{2850 - 178n}{17\sqrt{n}}\right) = \Phi\left(\frac{2850 - 178n}{17\sqrt{n}}\right).$$

We know from before that this is equivalent to $\Phi(3.719)$ and so we need to solve the following expression for n

$$\frac{2850 - 178n}{17\sqrt{n}} = 3.719 \implies 178n + 3.719 \times 17\sqrt{n} - 2850 = 0.$$

This is a quadratic with two real solutions, only one of which is positive, namely $n = 14.65$. Hence we will be safe that the raft will not sink with probability at least 0.9999 with 14 economists on the raft.

S1.5 What to expect when your expectation is expecting. Volume 1: Theory. Consider two discrete random variables X and Y that each admits a finite number of possible realizations, i.e. $X = \{x_1, x_2, \ldots, x_n\}$ and $Y = \{y_1, y_2, \ldots, y_m\}$. The joint probability function is given by $\mathbb{P}_{XY}(X = x_i, Y = y_j)$, $i = 1, \ldots, n$ and $j = 1, \ldots, m$. The corresponding marginal probabilities are $\mathbb{P}_X(X = x_i)$, $i = 1, \ldots, n$, and $\mathbb{P}_Y(Y = y_j)$, $j = 1, \ldots, m$. State the law of iterated expectations (LIE) for $\mathbb{E}[X]$ and the law of total probability (LTP) for $\mathbb{P}_X(X = x)$. Show that the law of iterated expectations follows from the law of total probability applied to the expectation $\mathbb{E}[X]$. Start by considering $\mathbb{E}[X] = \sum_{i=1}^{n} x_i \mathbb{P}_X(X = x_i)$ and then apply the law of total probability to $\mathbb{P}_X(X = x_i)$.

Solution: The law of iterated expectations is given by

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]] \tag{37}$$

Also, recall the definition of the expectation for a discrete random variable:

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \mathbb{P}_X(X = x_i) \tag{38}$$

In terms of the discrete random variables X, Y described in the question, the joint probability $\mathbb{P}_{XY}(\cdot)$, the marginal probabilities, $\mathbb{P}_X(\cdot)$ and $\mathbb{P}_Y(\cdot)$, and the conditional probability $\mathbb{P}_{X|Y}(\cdot)$, the law of total probability for some event $X = x$ is given by:

$$\mathbb{P}_X(X = x) = \sum_{j=1}^{m} \mathbb{P}_{XY}(X = x, Y = y_j) = \sum_{j=1}^{m} \mathbb{P}_{X|Y}(X = x \,|\, Y = y_j)\, \mathbb{P}_Y(Y = y_j),$$

where $x = \{x_1, \ldots, x_n\}$ are the possible realizations of X. Let's show that $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$ can be derived by applying the law of total probability to the probability $\mathbb{P}_X(X = x_i)$ inside the expectation of X:

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \mathbb{P}_X(X = x_i)$$

$$
\begin{aligned}
[\![\text{expand } \mathbb{P}_X(X = x_i) \text{ using LTP}]\!] &= \sum_{i=1}^{n} x_i \sum_{j=1}^{m} \mathbb{P}_{X|Y}(X = x_i \,|\, Y = y_j)\, \mathbb{P}_Y(Y = y_j) \\
[\![\text{swap order of summation}]\!] &= \sum_{j=1}^{m} \left[\sum_{i=1}^{n} x_i \mathbb{P}_{X|Y}(X = x_i \,|\, Y = y_j) \right] \mathbb{P}_Y(Y = y_j) \\
[\![Y_j \text{ fixed inside square bracket}]\!] &= \sum_{j=1}^{m} \mathbb{E}[X \,|\, Y = y_j]\, \mathbb{P}_Y(Y = y_j) \\
[\![\text{definition of expectation}]\!] &= \mathbb{E}[\mathbb{E}[X|Y]]
\end{aligned}
$$

The law of iterated expectations follows from the law of total probability applied to the expectation. The same can be shown in the opposite direction:

$$
\begin{aligned}
\mathbb{E}[\mathbb{E}[X|Y]] &= \mathbb{E}\left[\sum_{i=1}^{n} x_i \mathbb{P}_{X|Y}(X = x_i \,|\, Y) \right] \\
[\![\text{expand outer expectation}]\!] &= \sum_{j=1}^{m} \left[\sum_{i=1}^{n} x_i \mathbb{P}_{X|Y}(X = x_i \,|\, Y = y_j) \right] \mathbb{P}_Y(Y = y_j) \\
[\![\text{definition of conditional prob}]\!] &= \sum_{j=1}^{m} \sum_{i=1}^{n} x_i \mathbb{P}_{XY}(X = x_i, Y = y_j) \\
[\![\text{swap order of summation}]\!] &= \sum_{i=1}^{n} x_i \sum_{j=1}^{m} \mathbb{P}_{XY}(X = x_i, Y = y_j) \\
[\![\text{definition of marginal prob}]\!] &= \sum_{i=1}^{n} x_i \mathbb{P}_X(X = x_i) \\
[\![\text{definition of expectation}]\!] &= \mathbb{E}[X].
\end{aligned}
$$

Note that you can equivalently use the probability mass function notation, for example $f_{XY}(x, y) = \mathbb{P}_{XY}(X = x, Y = y)$, or $f_X(x) = \mathbb{P}_X(X = x)$.

S1.6 What to expect when your expectation is expecting. Volume 2: Practice. Consider the following joint distribution function for two discrete random variables, $\mathbb{P}_{XY}(X = x, Y = y)$, where $X = \{9, 10, 11\}$ and $Y = \{98, 99, 100, 101, 102\}$. The joint probability mass function is given in the following table:

X/Y	98	99	100	101	102	$\mathbb{P}_X(X = x)$
9	0.02	0.03	0.11	0.05	0.03	**0.24**
10	0.09	0.10	0.15	0.11	0.06	**0.51**
11	0.04	0.05	0.12	0.02	0.02	**0.25**
$\mathbb{P}_Y(Y = y)$	**0.15**	**0.18**	**0.38**	**0.18**	**0.11**	**1.00**

(a) Compute marginal distributions $\mathbb{P}_X(X = x)$ and $\mathbb{P}_Y(Y = y)$.

Solution: The answers are in the table above, where marginal distributions are computed using summations of the table entries over rows for $\mathbb{P}_Y$ and summations

over columns for $\mathbb{P}_X$, respectively, since

$$\mathbb{P}_X(X=x)=\sum_y \mathbb{P}_{XY}(X=x,Y=y)$$
$$\mathbb{P}_Y(Y=y)=\sum_x \mathbb{P}_{XY}(X=x,Y=y)$$

(b) Use marginal distributions to find unconditional expectations $\mathbb{E}[X]$ and $\mathbb{E}[Y]$.
Solution: Use the unconditional expectation formula

$$\mathbb{E}[X]=\sum_x x\mathbb{P}_X(X=x)=9\times 0.24+10\times 0.51+11\times 0.25=10.01$$
$$\mathbb{E}[Y]=\sum_y y\mathbb{P}_Y(Y=y)=99.92.$$

(c) Find conditional distribution functions $\mathbb{P}_{X|Y}(X=x\,|\,Y=y)$, for each value of y, and $\mathbb{P}_{Y|X}(Y=y\,|\,X=x)$ for each value of x. Present them in a table format.
Solution: Use the definition of conditional probability

$$\mathbb{P}_{X|Y}(X=x\,|\,Y=y)=\frac{\mathbb{P}_{XY}(X=x,Y=y)}{\mathbb{P}_Y(Y=y)}$$
$$\mathbb{P}_{Y|X}(Y=y\,|\,X=x)=\frac{\mathbb{P}_{XY}(X=x,Y=y)}{\mathbb{P}_X(Y=x)}$$

to compute the following conditional distributions:

$\mathbb{P}(X=x\,\vert\,Y=y)$	$Y=98$	$Y=99$	$Y=100$	$Y=101$	$Y=102$
$X=9$	0.13	0.17	0.29	0.28	0.27
$X=10$	0.60	0.56	0.39	0.61	0.55
$X=11$	0.27	0.28	0.32	0.11	0.18
$\mathbb{E}[X\vert Y=y]$	**10.13**	**10.11**	**10.03**	**9.83**	**9.91**

$\mathbb{P}(Y=y\,\vert\,X=x)$	$Y=98$	$Y=99$	$Y=100$	$Y=101$	$Y=102$	$\mathbb{E}[Y\vert X=x]$
$X=9$	0.08	0.13	0.46	0.21	0.13	**100.17**
$X=10$	0.18	0.20	0.29	0.22	0.12	**99.90**
$X=11$	0.16	0.20	0.48	0.08	0.08	**99.92**

(d) Find conditional expectations of X, $\mathbb{E}[X|Y=y]$, for each value of y. Hence, use the law of iterated expectations to compute $\mathbb{E}[X]$.
Solution: For a given value of $Y=y$, conditional probability of X is

$$\mathbb{E}[X|Y=y]=\sum_x x\mathbb{P}_{X|Y}(X=x\,|\,Y=y),$$

The solutions are shown in the table for $\mathbb{P}(X=x|Y=y)$ above, and an example of a computation is

$$\mathbb{E}[X|Y=98]=\sum_x xP_{X|Y}(X=x\,|\,Y=98)=9\times 0.13+10\times 0.6+11\times 0.27=10.1$$

The unconditional expectation of X can then be found using the LIE

$$\begin{aligned}\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]] &= \sum_y \mathbb{E}[X|Y=y]\,\mathbb{P}_Y(Y=y) \\ &= 10.13 \times 0.15 + 10.11 \times 0.18 + 10.03 \times 0.38 + \ldots \\ &= 10.01\end{aligned}$$

which of course agrees with the answer from (b).

(e) Find conditional expectations of Y, $\mathbb{E}[Y|X=x]$, for each value of x. Hence, use the law of iterated expectations to compute $\mathbb{E}[Y]$.

Solution: The conditional expectations are given in the table for $\mathbb{P}(Y=y|X=x)$ above. These are then used to compute unconditional expectation:

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]] = \sum_x \mathbb{E}[Y|X=x]\,\mathbb{P}_X(X=x) = 99.92.$$

I hope you are now convinced that the law of iterated expectations works!

S1.7 Dummy Wars. Episode IV: A new education.

We are interested in analyzing a population relationship between hourly wage (y_i) and the following binary (dummy) variables:

$$\texttt{uni}_i = \begin{cases} 1 & \text{university degree} \\ 0 & \text{no university degre} \end{cases} \qquad \texttt{male}_i = \begin{cases} 1 & \text{male} \\ 0 & \text{female} \end{cases}$$

In addition, define the variable $\texttt{fem}_i = 1 - \texttt{male}_i$.

(a) Consider the following linear regressions specifications for $i = 1, \ldots, n$:

$$y_i = \alpha + \beta \texttt{male}_i + u_i \tag{39}$$

$$y_i = \beta^M \texttt{male}_i + \beta^F \texttt{fem}_i + v_i \tag{40}$$

where u_i are v_i are iid error terms with zero mean and constant variance. Interpret the population coefficients α, β and β^M, β^F. Explain why the intercept is included in (39) but not in (40). Express α, β in terms of β^M, β^F. What is the relationship between the population error terms u_i and v_i?

Solution: The interpretation of the coefficients in the first regressions are:

$$\begin{aligned}\mathbb{E}[y_i \,|\, \texttt{male}_i = 0] &= \alpha \\ \mathbb{E}[y_i \,|\, \texttt{male}_i = 1] &= \alpha + \beta\end{aligned}$$

that is α is the expected wage for females and β is the expected additional wage for males compared to females, so that the expected wage for males is $\alpha + \beta$. The second regression has

$$\begin{aligned}\mathbb{E}[y_i \,|\, \texttt{male}_i = 1, \texttt{fem}_i = 0] &= \beta^M \\ \mathbb{E}[y_i \,|\, \texttt{male}_i = 0, \texttt{fem}_i = 1] &= \beta^F.\end{aligned}$$

The relationship between the coefficients is $\beta^F = \alpha$, $\beta^M = \alpha + \beta$, or equivalently $\beta = \beta^M - \beta^F$. Note that in (39) the default category is female ($\texttt{male}_i = 0$) and the effect is captured by the intercept. In (40) this default category enters the regression explicitly as an additional explanatory variable, $\texttt{fem}_i$. Hence, if we include an intercept α is regression (40) we will run into the problem of perfect multicollinearity, since $\texttt{fem}_i$ is a linear function of $\texttt{male}_i$. Finally, notice that the two population regressions are equivalent and hence the error terms are the same, $u_i = v_i$ for all i.

(b) How would you estimate the parameters α, β and β^M, β^F using a random sample of observations on wages, $y_1, \ldots, y_n$?

Solution: We can estimate the population parameters using their sample counterparts. Namely, since β^M and β^F are the expected wages (population mean wages) for males and females, we could use as their estimators the corresponding sample means for males, $\bar{y}^M$, and females, $\bar{y}^F$, respectively. Consequently, $\hat{\alpha} = \bar{y}^F$ and $\hat{\beta} = \bar{y}^F + \bar{y}^M$. Formally, if the sample is split such that the observations indexed $i = 1, \ldots, n_1$ represent males, and the observations indexed $i = n_1 + 1, \ldots n$ represent females (i.e. there are a total of n_1 males and $n - n_1$ females in the sample) the corresponding sub-sample means are given by

$$\bar{y}^M = \frac{1}{n_1} \sum_{i=1}^{n_1} y_i, \qquad \bar{y}^F = \frac{1}{n - n_1} \sum_{i=n_1+1}^{n} y_i. \tag{41}$$

(c) Consider the following regression

$$y_i = \alpha + \beta \texttt{male}_i + \delta \texttt{uni}_i + \gamma \left(\texttt{male}_i \times \texttt{uni}_i\right) + e_i, \quad i = 1, \ldots, n, \tag{42}$$

where e_i is an iid error term with zero mean and constant variance. Compute conditional expectations $\mathbb{E}\left[y_i | \texttt{male}_i, \texttt{uni}_i\right]$ for all possible values of $\texttt{male}_i$ and $\texttt{uni}_i$. Hence, interpret the population regression coefficients.

Solution: Interpretation follows directly from the conditional expectations:

$$\begin{aligned}
\mathbb{E}\left[y_i \,|\, \texttt{male}_i = 0, \texttt{uni} = 0\right] &= \alpha \\
\mathbb{E}\left[y_i \,|\, \texttt{male}_i = 1, \texttt{uni} = 0\right] &= \alpha + \beta \\
\mathbb{E}\left[y_i \,|\, \texttt{male}_i = 0, \texttt{uni} = 1\right] &= \alpha + \delta \\
\mathbb{E}\left[y_i \,|\, \texttt{male}_i = 1, \texttt{uni} = 1\right] &= \alpha + \beta + \delta + \gamma
\end{aligned}$$

So that α is the expected wage for a female without a degree; β is the additional expected wage for a male without a degree compared to a female without a degree (gender effect for non-graduates); δ is is the additional expected wage for a female with a degree compared to a female without a degree (degree effect for females); $\delta + \gamma$ is the additional expected wage for males with a degree compared to males without a degree (interaction of gender and degree effects).

(d) Let us define a new categorical variable for the educational attainment level measured in years of schooling completed, $\texttt{educ}_i = \{0, 1, 2, \ldots 18\}$. Furthermore, define a set of 19 dummy variables for each educational level from 0 to 18:

$$\texttt{educ_j} = \begin{cases} 1 & \text{if } \texttt{educ}_i = j \\ 0 & \text{otherwise} \end{cases}$$

where $j = 1, \ldots, 18$ represents years of education completed. For example, $\texttt{educ_12}_i$ is a dummy variable that takes the value of 1 for those individuals who have attained exactly 12 years of education ($\texttt{educ}_i = 12$) and zero for all other individuals. Consider the following regressions

$$y_i = \alpha + \beta \texttt{male}_i + \delta \texttt{educ}_i + u_i \tag{43}$$

$$y_i = \alpha + \beta \texttt{male}_i + \delta_1 \texttt{educ_1}_i + \cdots + \delta_{18} \texttt{educ_18}_i + v_i \tag{44}$$

where u_i and v_i are iid errors terms with zero mean and constant variance. The latter regression can be written in concise notation as

$$y_i = \alpha + \beta \texttt{male}_i + \sum_{j=1}^{18} \delta_j \texttt{educ_j}_i + v_i.$$

Interpret the coefficient δ from (43) and the coefficients δ_j, $j = 1, \ldots, 18$, from (44). Explain why regression (44) allows for non-linear effects of additional years of education on wage, while regression (43) does not. How will the residuals u_i from (43) compare with residuals v_i from (44)?

Solution: The slope coefficient in (43) is the expected additional wage for an extra year of education, given the person's gender:

$$\delta = \mathbb{E}\left[y_i \,|\, \texttt{male}_i, \texttt{educ}_i = j + 1\right] - \mathbb{E}\left[y_i \,|\, \texttt{male}_i, \texttt{educ}_i = j\right].$$

Note that here we use the conditional expectation interpretation rather than the partial derivative of y_i with respect to $\texttt{educ}_i$, since the explanatory variable is not continuous, so the derivative does not make much sense in this case. The effect δ is the same for all levels of education j. This means that moving from 1 year of education to 2 years will yield the same expected increase in wage as moving from 15 years of education to 16. This not very realistic, as the effects most probably differ depending on the current level of education, making the effect non-linear - the effect of an increase from 1 to 2 should be different from the effect of an increase from 15 to 16. Therefore regression (43) is most probably misspecified, as it is not flexible enough to show the full variability of effects.

Regression (44), on the other hand, corrects this drawback and allows for non-linearity in effects to be expressed explicitly. Namely, t'[;he coefficients δ_j shows the additional effect on wages of having j years of education compared to 0 years of education:

$$\delta_j = \mathbb{E}\left[y_i \,|\, \texttt{male}_i, \texttt{educ}_i = j\right] - \mathbb{E}\left[y_i \,|\, \texttt{male}_i, \texttt{educ}_i = 0\right]$$

Note that the default category is $\texttt{educ_0}_i$, that is when $\texttt{educ}_i = 0$. All δ_j coefficients are compared against the default effect of $\texttt{educ_0}_i$ that is measured by the intercept α. Recall from (a) that the variable representing the default category is not included in the regression explicitly, as this will lead to perfect multicollinearity. As an example, the additional effect of studying 12 years as compared to having no education is

$$\delta_{12} = \mathbb{E}\left[y_i \,|\, \texttt{male}_i, \texttt{educ}_i = 12\right] - \mathbb{E}\left[y_i \,|\, \texttt{male}_i, \texttt{educ}_i = 0\right]$$

Expressed in terms of notation of regression (44) with the 18 distinct dummy variables for the educational levels, this can be written more formally as

$$\begin{aligned}\delta_j = &\mathbb{E}\left[y_i \,|\, \texttt{male}_i, \texttt{educ_1} = 0, \ldots, \texttt{educ_12} = 1, \texttt{educ_13} = 0, \ldots\right] \\ &- \mathbb{E}\left[y_i \,|\, \texttt{male}_i, \texttt{educ_1} = 0, \ldots, \texttt{educ_12} = 0, \texttt{educ_13} = 0, \ldots\right]\end{aligned}$$

where the top part is the expected wage, given gender, for someone with 12 years of education, that is when $\texttt{educ_12} = 1$ and all other dummies equal to zero, and the bottom part is the expected wage for someone with 0 years of education. It is realistic to expect that the magnitude of these effects is not constant and so $\delta_1 \neq \delta_2 \neq \ldots \neq \delta_{18}$. Therefore, δ_j from (44) allow to distinguish between the effects of an additional year of education for different current attainment levels, while δ from (43) shows the average extra wage effect of an additional year of education, irrespective of the current attainment level.

For instance, the expected wage effect of an extra year of education for someone who currently has 11 years of education predicted by regression (44) is $\delta_{12} - \delta_{11}$ (that is increasing by one year from 11 to 12). The effect of moving from 1 year to 2 is given by $\delta_2 - \delta_1$. Sadly, regression (43) would predict that both these effects are equal to δ. It is clear that regression (44) is a better model than regression (43), that is has better explanatory power, and hence the residuals v_i are likely to be smaller than u_i.